

URBANCART ANALYTICS — EXECUTIVE SUMMARY

Context: UrbanCart's operational data was previously siloed across SQLite transaction records, legacy CRM exports, and supplier catalogs. This document summarizes the outcomes of our data integration, cleaning, and custom analytical pipeline deployed to guide business expansion.

Key Finding 1: Champions Drive Revenue

Our custom RFM segmentation model identified that the 'Champions' segment (high-value, highly frequent purchasers) represents only 16.3% of the customer base but generates over 50% (\$5.39M) of total net revenue. Retaining this group is paramount.

Key Finding 2: Consistent Revenue Expansion

Monthly revenue grew from \$176k in January 2023 to over \$302k in December 2024. A linear model fit via the normal equation indicates a growth rate of \$7.11k per month, explaining 84.93% of the variance ($R^2 = 0.8493$). Revenue for January 2025 is projected to hit \$311.6k.

Key Finding 3: Books & Media Category Has Highest Profit Margins

Accounting for returns, discounts, and costs, 'Books & Media' achieves the highest profit margin at 50.1%, while 'Electronics' has the lowest margin at 38.6% due to high return rates and wholesale costs.

Key Finding 4: Price Typographical Errors Mangle Raw Insights

keystroke errors in products (e.g. a book priced at \$34,872.00 instead of \$25.00) inflated raw revenues. Implementing an IQR cleaning filter capped these outliers, correcting historical sales metrics by 10.4% and restoring data credibility.

Key Finding 5: High Mobile Conversion in Europe

Although sessions on mobile and desktop are similarly engaged, mobile users in Germany and France show an outstanding purchase conversion rate of 68.2%, signifying a major demographic marketing opportunity.

Recommended Actions:

- VIP Retention Campaigns:** Target the Champions segment with early access and premium loyalty rewards.
- Implement Safety Stock Levels:** Set warehouse restock alerts at the simulated reorder points (7.3 units for product 136).
- Launch Mobile-First Checkout:** Optimize checkout flows in European markets to capitalize on high mobile conversions.